BABS DIGITAL INNOVATIONS & LEARNING PLATFORM (BDIP)

Data Science & Analytics

Transform data into interpretable evidence for decisions.

Purpose

The BDIP Data Science domain provides methods for preparing, exploring, analysing, modelling, and communicating data. It supports scientific studies, organizational planning, monitoring, forecasting, and decision support while emphasizing data quality, provenance, uncertainty, privacy, and appropriate interpretation.

Core capabilities

- Data collection, validation, cleaning, and integration
- Descriptive and inferential statistics
- Exploratory analysis, visualization, and dashboards
- Statistical modelling, classification, and clustering
- Forecasting, time-series, geospatial, and uncertainty analysis
- Optimization and evidence-informed decision analytics

Operating workflow

Define the decision need → assess data quality → explore and model → validate results → communicate uncertainty → support action.

Expected contribution

Clearer patterns, reproducible analyses, measurable indicators, responsible predictions, and better-informed decisions.

Connections within BDIP

Research datasets; Knowledge management; AI and machine learning; Applications and dashboards; Governance, privacy, and data quality.